SUMMARY

- The Laplace transformation converts waveforms in the time domain to transforms in the s domain. The inverse transformation converts transforms into causal waveforms. A transform pair is unique if and only if $f(t)$ is causal.
- The Laplace transforms of basic signals like the step function, exponential, and sinusoid are easily derived from the integral definition. Other transform pairs can be derived using basic signal transforms and the uniqueness, linearity, time integration, time differentiation, and translation properties of the Laplace transformation.
- Proper rational functions with simple poles can be expanded by partial fraction to obtain inverse Laplace transforms. Simple real poles lead to exponential waveforms and simple complex poles to damped sinusoids. Partial-fraction expansions of improper rational functions and functions with multiple poles require special treatment.
- Using Laplace transforms to find the response of a linear circuit involves transforming the circuit differential equation into the s domain, algebraically solving for the response transform, and performing the inverse transformation to obtain the response waveform.
- The initial and final value properties determine the initial and final values of a waveform $f(t)$ from the value of $sF(s)$ at $s \to \infty$ and $s = 0$, respectively. The initial value property applies if $F(s)$ is a proper rational function. The final value property applies if all of the poles of $sF(s)$ are in the left half plane.

PROBLEMS

OBJECTIVE 9–1 LAPLACE TRANSFORM (SECTS. 9–1, 9–2, 9–3)

Find the Laplace transform of a given signal waveform using transform properties and pairs, or using the integral definition of the Laplace transformation. Locate the poles and zeros of the transform and construct a pole-zero diagram.

See Examples 9–1, 9–2, 9–3, 9–4, 9–5, 9–6, 9–7, 9–8, 9–9, 9–10

9–1 Find the Laplace transform of $f(t) = A[e^{-\alpha t} - 2e^{-\gamma t}]u(t)$. Locate the poles and zeros of $F(s)$.

9–2 Find the Laplace transform of $f(t) = A[(2 - \alpha t)e^{-\alpha t}]u(t)$. Locate the poles and zeros of $F(s)$.

9–3 Find the Laplace transform of $f(t) = A[1 - 2\cos(\beta t)]u(t)$. Locate the poles and zeros of $F(s)$.

9–4 Find the Laplace transform of $f(t) = A[\cos(\beta t) - \sin(\beta t)]u(t)$. Locate the poles and zeros of $F(s)$.

9–5 Find the Laplace transform of $f(t) = A\delta(t) - 2A\beta e^{-\beta t}\cos(\beta t)u(t)$. Locate the poles and zeros of $F(s)$.

9–6 Find the Laplace transform of $f(t) = A[2 - \alpha t - 2e^{-\alpha t}]u(t)$ for $\alpha > 0$. Locate the poles and zeros of $F(s)$.

9–7 Find the Laplace transforms of the following waveforms and plot their pole-zero diagrams:

(a) $f_1(t) = [5e^{-5t} - 10e^{-20t}]u(t)$
(b) $f_2(t) = [5\cos(10t) + 7\cos(20t)]u(t)$

9–8 Find the Laplace transforms of the following waveforms and plot their pole-zero diagrams.

(a) $f_1(t) = 3\delta(t) + [10e^{-10t} - 40e^{-40t}]u(t)$
(b) $f_2(t) = [20 - 15\cos(500t)]u(t)$

9–9 Find the Laplace transform of the following waveform. Locate the poles and zeros of $F(s)$.

(a) $f_1(t) = \delta(t) - (625te^{-50t})u(t)$
(b) $f_2(t) = [5 + e^{-20t} - 6\cos(10t) + 2\sin(10t)]u(t)$

9–10 Find the Laplace transforms of the following waveforms:

(a) $f_1(t) = 2\delta(t - 2)$
(b) $f_2(t) = e^{-50(t-1)}u(t - 1)$
(c) $f_3(t) = e^{-50(t-2)}u(t - 2)$

9–15 Find the Laplace transform of the following function. Locate the poles and zeros of $F(s)$.

$$f(t) = [1600t + 8 + 75e^{-10t} + 17e^{-50t}]u(t)$$

OBJECTIVE 9–2 INVERSE TRANSFORMS (SECTS. 9–4, 9–5)

Find the inverse transform of a given Laplace transform using partial-fraction expansion or basic transform properties and pairs.

See Examples 9–11, 9–12, 9–13, 9–14 and Exercises 9–7, 9–8, 9–9, 9–10, 9–11, 9–12, 9–13, 9–14

9–16 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \dfrac{s + 20}{s(s + 10)}$

(b) $F_2(s) = \dfrac{s^2 + 10s + 10}{s(s + 10)}$

9–17 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \dfrac{2(s + 5)}{s(s + 10)}$

(b) $F_2(s) = \dfrac{s^2}{(s + 5)(s + 10)}$

9–18 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \dfrac{20(s + 20)}{(s + 10)^2 + 400}$

(b) $F_2(s) = \dfrac{20(s - 20)}{(s + 10)^2 + 400}$

9–19 Find the inverse Laplace transforms of the following functions and sketch their waveforms for $\beta > 0$:

(a) $F_1(s) = \dfrac{\beta(s + \beta)}{s(s^2 + \beta^2)}$

(b) $F_2(s) = \dfrac{s(s + \beta)}{s^2 + \beta^2}$

9–20 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \dfrac{\alpha^2}{s^2(s + \alpha)}$

(b) $F_2(s) = \dfrac{\alpha^2}{s(s + \alpha)^2}$

9–21 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \dfrac{6\alpha^2}{(s + \alpha)(s + 2\alpha)(s + 4\alpha)}$

(b) $F_2(s) = \dfrac{6(s + 2\alpha)}{(s + \alpha)(s + 4\alpha)}$

9–22 Find the inverse transforms of the following functions:

(a) $F_1(s) = \dfrac{16}{(s + 2)(s^2 + 12s + 36)}$

(b) $F_2(s) = \dfrac{2(s^2 + 2)}{s(s^2 + 4)}$

9–23 Find the inverse transforms of the following functions:

(a) $F_1(s) = \dfrac{(s + 4)(s + 8)}{s(s + 2)(s + 6)}$

(b) $F_2(s) = \dfrac{3s^4 + 10s^2 + 4}{s(s^2 + 1)(s^2 + 4)}$

9–24 Find the inverse transforms for the following functions:

(a) $F_1(s) = \dfrac{30(s + 2)}{s(s^2 + 4s + 5)}$

(b) $F_2(s) = \dfrac{2s}{(s + 4)(s^2 + 4s + 8)}$

9–25 Find the inverse transforms for the following functions:

(a) $F_1(s) = \dfrac{2(s^2 + 16)}{s(s^2 + 8s + 32)}$

(b) $F_2(s) = \dfrac{s^2 + 30s + 800}{s(s^2 + 50s + 400)}$

9–26 Find the inverse transforms of the following functions:

(a) $F_1(s) = \dfrac{(s + 40)^2}{(s + 10)^2(s + 100)}$

(b) $F_2(s) = \dfrac{(s + 10)^2}{(s + 40)^2(s + 100)}$

9–27 A certain transform has a simple pole at $s = -20$, a simple zero at $s = -\gamma$, and a scale factor of $K = 1$. Select values for γ so the inverse transform is

(a) $f(t) = \delta(t) - 5e^{-20t}$
(b) $f(t) = \delta(t)$
(c) $f(t) = \delta(t) + 5e^{-20t}$

9–28 Find the inverse transforms of the following functions:

(a) $F_1(s) = \dfrac{(s + 40)e^{-2s}}{(s + 10)(s + 20)}$

(b) $F_2(s) = \dfrac{se^{-2s} + 20}{(s + 10)(s + 40)}$

9–29 Use Mathcad or MATLAB to find the inverse transform of the following function:

$$F(s) = \frac{s(s^2 + 3s + 4)}{(s + 2)(s^3 + 6s^2 + 16s + 16)}$$

9–30 Use Mathcad or MATLAB to find the inverse transform of the following function:

$$F(s) = \frac{40(s^3 + 2s^2 + s + 2)}{s(s^3 + 4s^2 + 4s + 16)}$$

PROBLEMS

OBJECTIVE 10–1 EQUIVALENT IMPEDANCE (SECTS. 10–1, 10–2)

Given a linear circuit, use series and parallel equivalences to find the poles and zeros of the equivalent impedance at specified terminal pairs. Select element values to obtain specified pole locations.
See Examples 10–2, 10–4, 10–5, 10–6 and Exercise 10–1, 10–2

10–1 Find $Z_{EQ}(s)$ in Figure P10–1. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

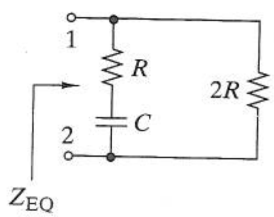

FIGURE P10–1

10–2 Find $Z_{EQ}(s)$ in Figure P10–2. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

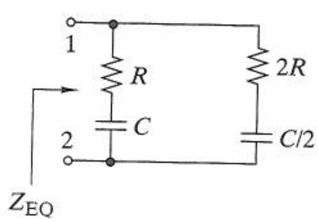

FIGURE P10–2

10–3 Find $Z_{EQ}(s)$ in Figure P10–3. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros for $R = 4\ \text{k}\Omega$, $L = 0.4$ H, and $C = 100$ nF.

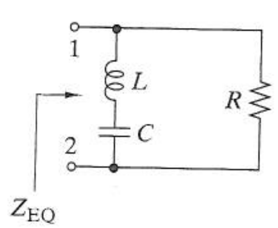

FIGURE P10–3

10–4 Find $Z_{EQ}(s)$ in Figure P10–4. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

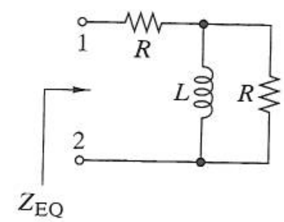

FIGURE P10–4

10–5 Find $Z_{EQ}(s)$ in Figure P10–5. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

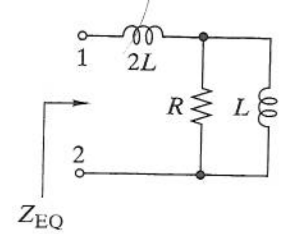

FIGURE P10–5

10–6 Find the poles and zeros of Z_{EQ} in Figure P10–6 for $R = 1\ \text{k}\Omega$, $L = 1$ H, and $C = 500$ nF.

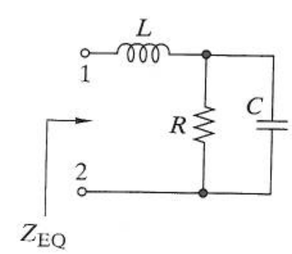

FIGURE P10–6

10–7 Find $Z_{EQ}(s)$ in Figure P10–7. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

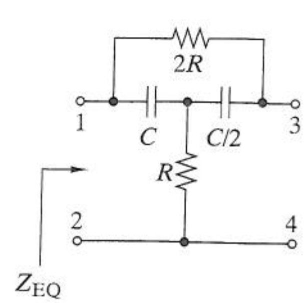

FIGURE P10–7

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10–11 The switch in Figure P10–11 has been in position A for a long time and is moved to position B at $t = 0$. Transform the circuit into the s domain and solve for $I_L(s)$ and $i_L(t)$ in symbolic form.

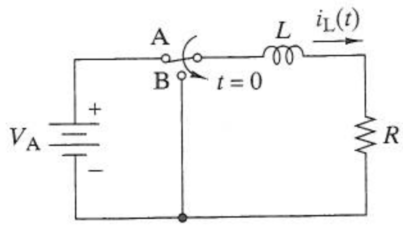

FIGURE P10–11

10–12 The switch in Figure P10–11 has been in position B for a long time and is moved to position A at $t = 0$. Transform the circuit into the s domain and solve for $I_L(s)$ and $i_L(t)$ in symbolic form.

10–13 The switch in Figure P10–13 has been in position A for a long time and is moved to position B at $t = 0$. Transform the circuit into the s domain and solve for $I_C(s)$ and $i_C(t)$ in symbolic form.

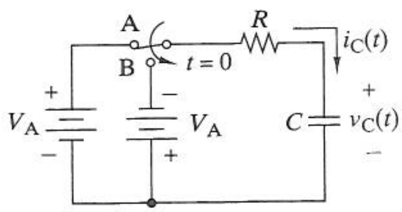

FIGURE P10–13

10–14 The switch in Figure P10–13 has been in position B for a long time and is moved to position A at $t = 0$. Transform the circuit into the s domain and solve for $V_C(s)$ and $v_C(t)$ in symbolic form.

10–15 There is no initial energy stored in the circuit in Figure P10–15. Transform the circuit into the s domain and find $I_L(s)$ and $i_L(t)$ when $v_1(t) = V_A\, u(t)$.

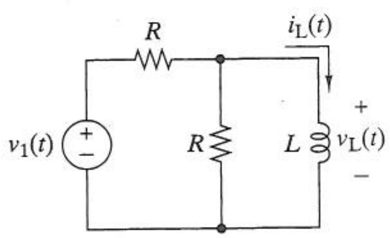

FIGURE P10–15

10–16 There is no initial energy stored in the circuit in Figure P10–15. Transform the circuit into the s domain and find $V_L(s)$ and $v_L(t)$ when $v_1(t) = V_A u(t)$.

10–17 The switch in Figure P10–17 has been in position A for a long time and is moved to position B at $t = 0$.

(a) Transform the circuit into the s domain and solve for $I_L(s)$ in symbolic form.

(b) Find $i_L(t)$ for $R_1 = R_2 = 500\ \Omega$, $L = 250$ mH, $C = 4\ \mu$F, and $V_A = 15$ V.

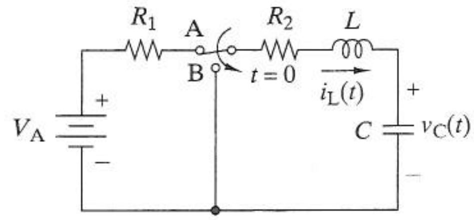

FIGURE P10–17

10–18 The switch in Figure P10–17 has been in position B for a long time and is moved to position A at $t = 0$.

(a) Transform the circuit into the s domain and solve for $V_C(s)$ in symbolic form.

(b) Find $v_C(t)$ for $R_1 = R_2 = 500\ \Omega$, $L = 500$ mH, $C = 1\ \mu$F, and $V_A = 15$ V.

10–19 The circuit in Figure P10–19 is in the zero state. Find the s-domain relationship between the input $I_1(s)$ and the output $I_R(s)$.

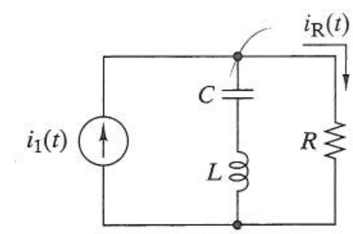

FIGURE P10–19

10–20 The circuit in Figure P10–20 is in the zero state. Find the s-domain relationship between the input $I_1(s)$ and the output $V_O(s)$.

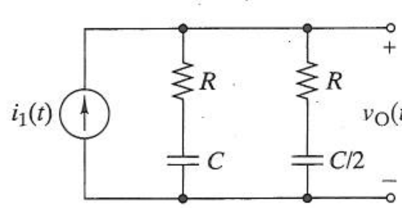

FIGURE P10–20

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10–27 The circuit in Figure P10–27 is in the zero state. Find the Thévenin equivalent to the left of the interface.

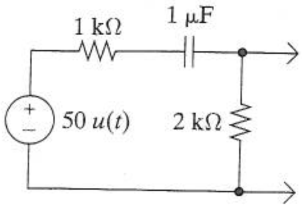

FIGURE P10–27

10–28 There is no initial energy stored in the circuit in Figure P10–28. Transform the circuit into the s domain and use superposition to find $V(s)$. Identify the forced and natural poles in $V(s)$.

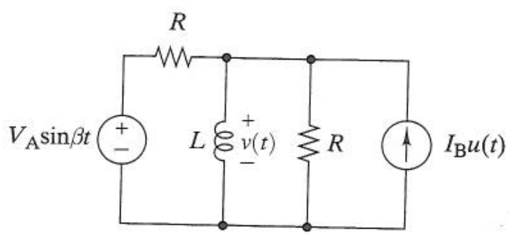

FIGURE P10–28

10–29 The equivalent impedance between a pair of terminals is

$$Z(s) = 1000\left[\frac{s + 2000}{s + 1000}\right]$$

A voltage $v(t) = 10u(t)$ is applied across the terminals. Find the resulting current response $i(t)$.

10–30 There is no initial energy stored in the circuit in Figure P10–30. Use circuit reduction to find the output voltage $V_2(s)$ in terms of the input voltage $V_1(s)$.

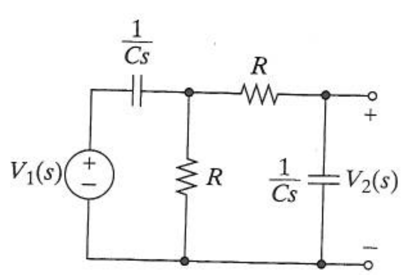

FIGURE P10–30

OBJECTIVE 10–3 GENERAL CIRCUIT ANALYSIS (SECTS. 10–4, 10–5, 10–6)

Given a linear circuit:

(a) Determine the initial conditions (if not given) and transform the circuit into the s domain.
(b) Solve for zero-state and/or zero-input response transforms and waveforms using node-voltage or mesh-current methods.
(c) Identify the forced and natural poles or select circuit parameters to place the natural poles at specified locations.

See Examples 10–10, 10–11, 10–13, 10–14, 10–15, 10–16, 10–17

10–31 There is no initial energy stored in the circuit in Figure P10–31.

(a) Transform the circuit into the s domain and formulate mesh-current equations.
(b) Solve these equations for $I_2(s)$ in symbolic form.
(c) Find $i_2(t)$ for $v_1(t) = 100u(t)$ V, $R_1 = 1$ kΩ, $R_2 = 2$ kΩ, $L = 4$H, and $C = 500$ nF.

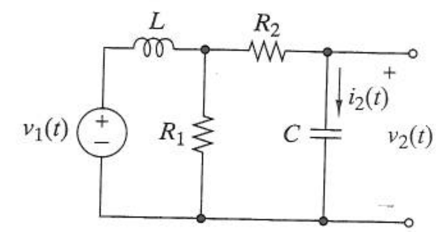

FIGURE P10–31

10–32 There is no initial energy stored in the circuit in Figure P10–31.

(a) Transform the circuit into the s domain and formulate node-voltage equations.
(b) Solve these equations for $V_2(s)$ in symbolic form.
(c) Find $v_2(t)$ for $v_1(t) = 10u(t)$ V, $R_1 = R_2 = 500$ Ω, $L = 0.5$ H, and $C = 2$ μF.

10–33 There is no initial energy stored in the circuit in Figure P10–33.

(a) Transform the circuit into the s domain and formulate node-voltage equations.
(b) Solve these equations for $V_2(s)$ in symbolic form.
(c) Find $v_2(t)$ for $v_1(t) = 30u(t)$ V, $R_1 = 10$ kΩ, $R_2 = 20$ kΩ, $C_1 = 1$ μF, and $C_2 = 0.5$ μF.